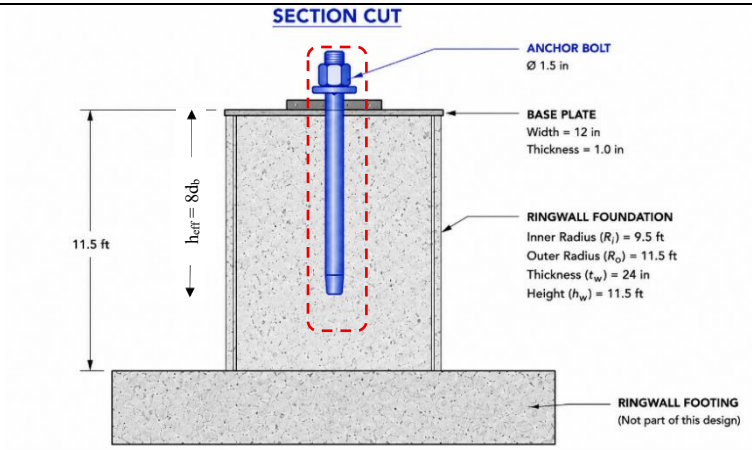
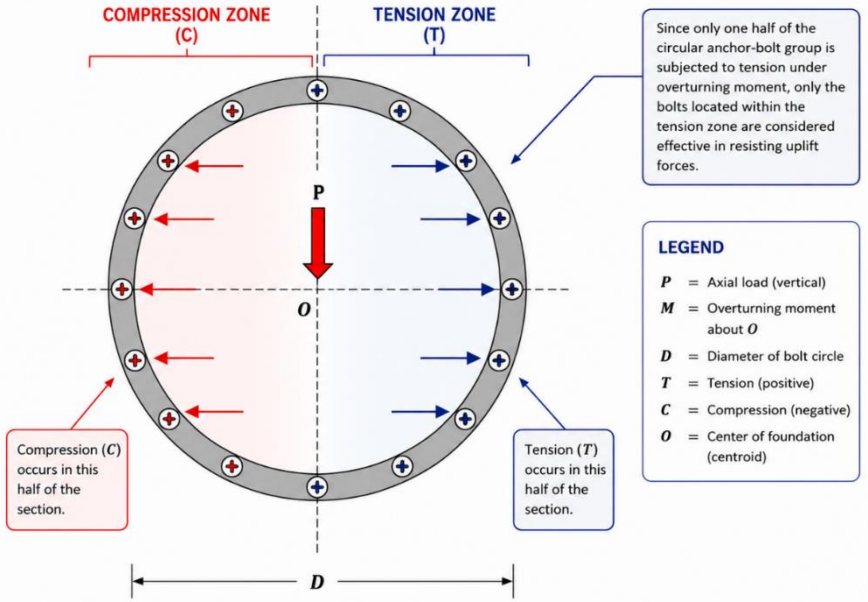
1. Anchor Bolts segment

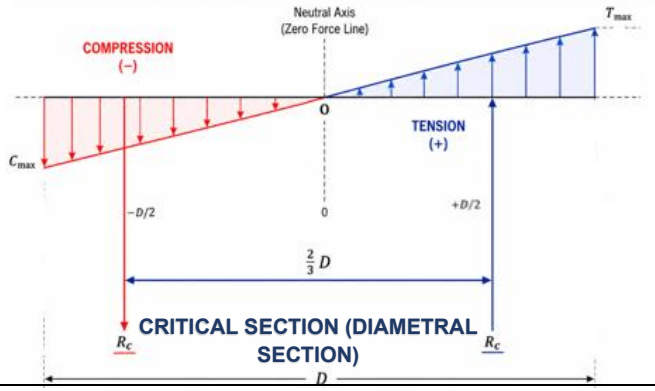
Input

Design Parameter	Symbol	Value	Source
Number of anchor bolts	N_β	36.0	User Input
Anchor bolt diameter	d_β	1.5 in	User Input
Hole diameter	d_h	1.625 in	User Input
Bolt circle radius	r_β	150.0 in	Calculated / input
Steel yield strength	F_γ	36 ksi	User Input
Steel ultimate strength	F_u	58 ksi	User Input
Factored axial load (LRFD)	P_u	3,895.2 kips	Main Tank Design (Predefined)
Resistance factor	ϕ	0.75	Design input
Factored moment (LRFD)	M_u	5,640.2 kip-ft	Main Tank Design (Predefined)
Bolt circle diameter	D	25.0 ft	Main Tank Design (Predefined)
Total shear force (LRFD)	V_u	86.4 kips	Main Tank Design (Predefined)
Edge distance	e	3.0 in	User Input
Concrete compressive strength	f'c	4,000 psi	User Input
Effective embedment depth	h_ef	25.0 in	User Input
Number of segments	n_s	4.0	User Input
Angular spacing in number of segment	θ_i	10.0	User Input
Segment number	i	2.0	User Input

- **N_b** = total number of anchor bolts
- **db** = nominal anchor bolt diameter (in)
- **dh** = anchor bolt hole diameter (in)
- **rb** = anchor bolt circle radius (ft or in)
- **αb** = start angle of first anchor bolt (deg)
- **θ_{seg}** = segment angle (deg), if bolts are checked by segment
- **n_s** = number of segments
- **t_{bp}** = base plate thickness (in)
- **f_y** = bolt yield strength
- **f_u** = bolt ultimate strength
- **T_u** = design uplift/tension demand on one bolt or bolt group
  - **V_u** = design shear demand on one bolt or bolt group
  - **φ** = resistance factor, if your software
  - includes strength check
  - **e** = edge distance from hole to plate edge
  - **s** = bolt spacing, if user enters directly
  - **nbs** = number of bolts per segment







Given

Tank radius: $R = 12'6" \; ft$

Governing moment: $M_u = 5640.178 \; kip - \text{ft} ft$,

Inner face: 11.5−1=10.5ft (matches9′−6")

Outer face: 13.5+1=12.5ft

Ring wall thickness = 2 ft

1. Governing Overturning Moment: From calculation

$$\boldsymbol{M_u = 5640.178 \text{ kip-ft}}$$

Diameter of bolt circle:

$$D = 25 \text{ ft}$$

Effective lever arm:

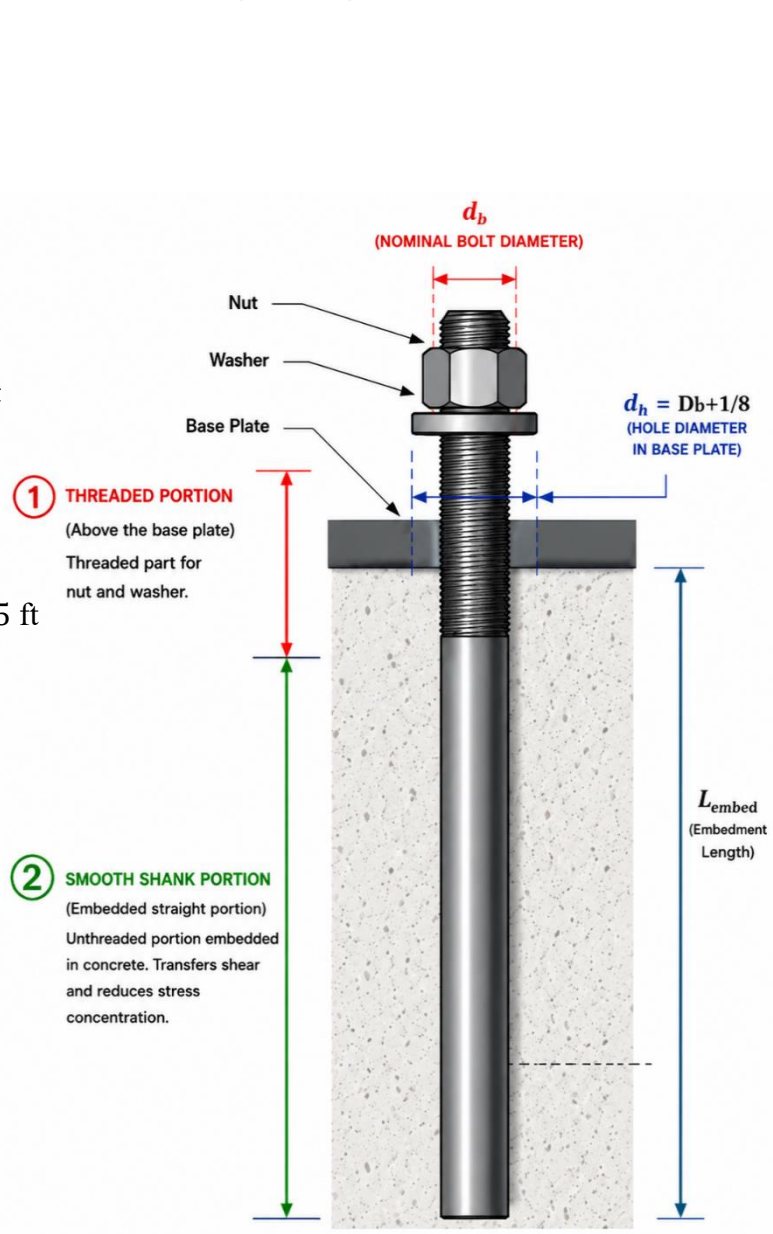
$$0.67D = 0.67(25) = 16.75 \text{ ft}$$

Total uplift force:

$$T_u = \frac{M_u}{0.67D}$$

$$T_u = \frac{5640.2}{16.75}$$

$$\boldsymbol{T_u = 336.73 \text{ kips}}$$



2. Tension Force Per Bolt

You used:

$$T_{bolt} = \frac{2T}{N_b}$$

This is correct for a simplified circular anchor group assumption.

Given:

$N_b = 36$

$$T_{bolt} = \frac{2(336.73)}{36}$$

$$T_{bolt} = \mathbf{18.71\ kips}$$

3. Bolt Area

We choose reasonable trial values:

- $N_b = 20$ bolts
- $d_b = 1.5\ in$….. (In accordance with ACI 318 anchor design provisions

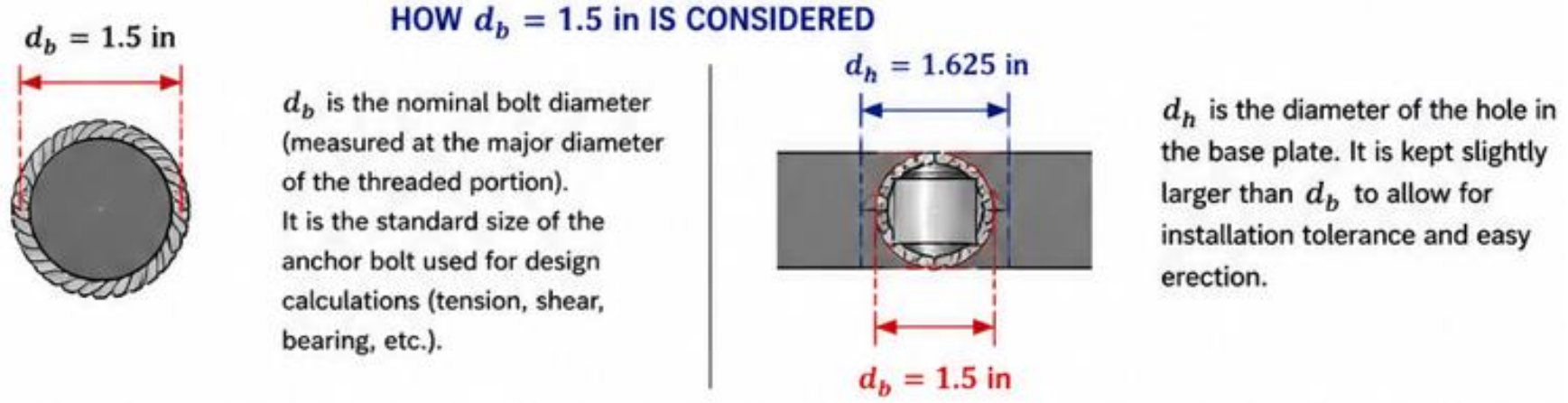
the anchor-bolt diameter d_b is taken as the nominal bolt diameter, corresponding to the major diameter of the threaded anchor rod.)

- $d_h = 1.625\ in$ (1/8 + d_b)
- $r_b = 10.75\ ft = 129\ in$
- n= 6 threads/in…….(ASME B1.1 — Unified Inch Screw Threads (UN and UNR Thread Form))
- $\alpha_b = 0°$

$$A_b = \frac{\pi d_b^2}{4}$$

$$A_b = \frac{\pi(1.5)^2}{4}$$

$$\boldsymbol{A_b = 1.767\ in^2}$$



Bolt Tensile Stress Area (Recommended)
For threaded anchor rods, ACI 318 and ASTM anchor design practice recommend using the tensile stress area rather than gross shank area because threaded portions govern tensile rupture.
Use:

$$A_t = 0.7854\left(d_b - \frac{0.9743}{n}\right)^2$$

Where:

- A_t= tensile stress area
- d_b= nominal bolt diameter
- n= threads per inch

For 1.5 in diameter anchor rod:

$$A_t \approx 1.405 \text{ in}^2$$

Adopt:

$$A_t = 1.41 \text{ in}^2$$

instead of gross area:

$$\boldsymbol{A_b = 1.767 \text{ in}^2}$$

4. Hole Area

$$A_h = \frac{\pi d_h^2}{4}$$
$$A_h = \frac{\pi(1.625)^2}{4}$$
$$\boldsymbol{A_h = 2.074 \text{ in}^2}$$

5. Angular Spacing

$$\Delta\theta_b = \frac{360}{N_b}$$
$$\boldsymbol{\frac{360}{36} = 10^\circ}$$

6. Bolt Coordinates
For bolt i:

$$\theta_i = (i-1) \times 10^\circ$$
$$x_i = r_b \cos\theta_i$$

$$y_i = r_b \sin\theta_i$$

e.g. if i=1 xi=0, Where: $r_b = 10.75\ ft$

If i=2

$$\theta_i = (2-1) \times 10° = 10°$$

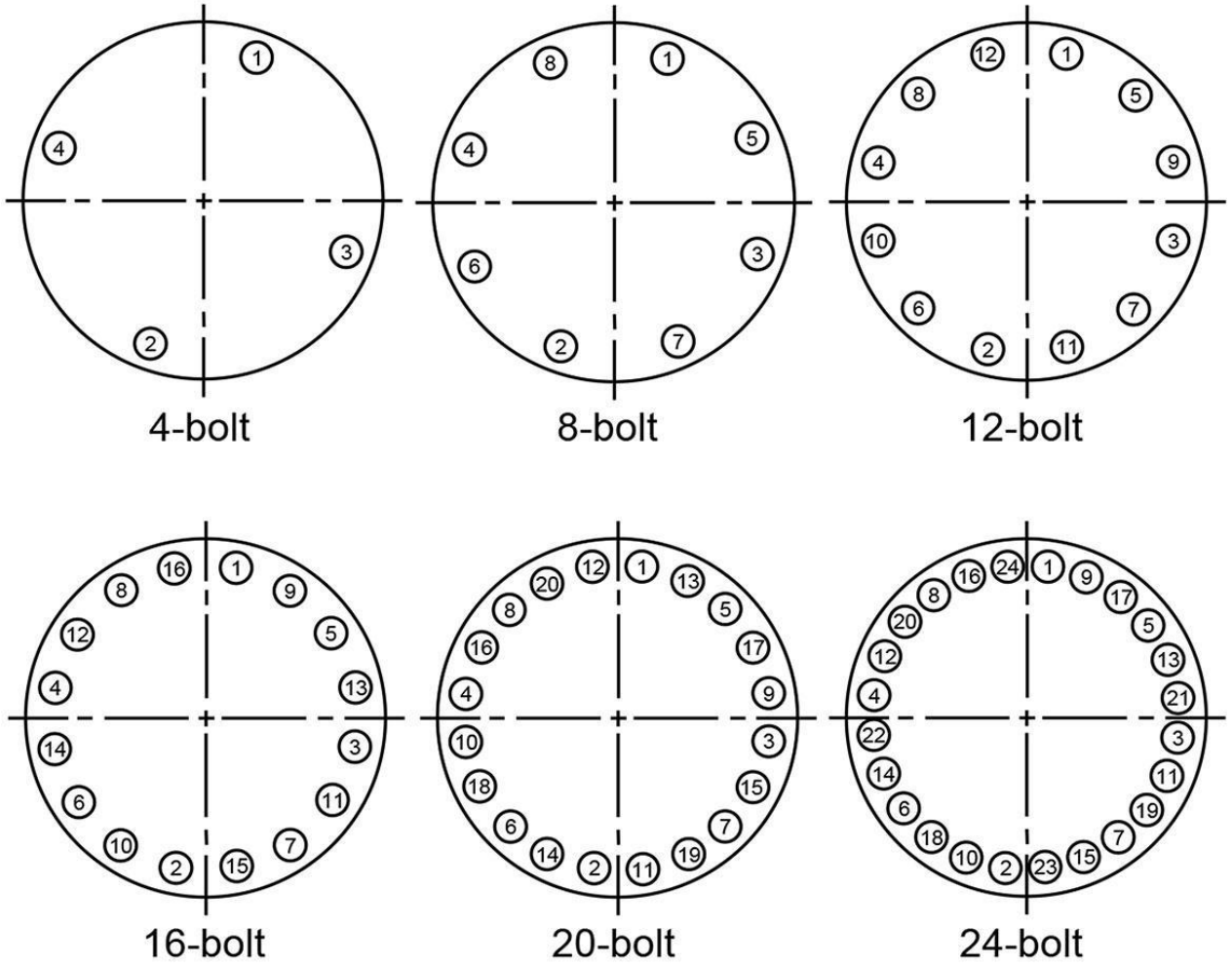
$$x_2 = 10.75 \text{ ft x } \cos 18° = 12.31\ ft$$

$$y_2 = 10.75 \text{ ft x } \sin 18° = 2.17\ ft$$

Anchor Bolt Layout Table

Bolt	Angle (°)	x (ft)	y (ft)	Location
1	0	12.50	0.00	Right
2	10	12.31	2.17	Upper-right
3	20	11.75	4.28	Upper-right
4	30	10.83	6.25	Upper-right
5	40	9.58	8.03	Upper-right
6	50	8.03	9.58	Upper-right
7	60	6.25	10.83	Upper-right
8	70	4.28	11.75	Upper-right
9	80	2.17	12.31	Upper-right
10	90	0.00	12.50	Top
11	100	-2.17	12.31	Upper-left
12	110	-4.28	11.75	Upper-left
13	120	-6.25	10.83	Upper-left
14	130	-8.03	9.58	Upper-left
15	140	-9.58	8.03	Upper-left
16	150	-10.83	6.25	Upper-left
17	160	-11.75	4.28	Upper-left
18	170	-12.31	2.17	Upper-left
19	180	-12.50	0.00	Left
20	190	-12.31	-2.17	Lower-left
21	200	-11.75	-4.28	Lower-left
22	210	-10.83	-6.25	Lower-left
23	220	-9.58	-8.03	Lower-left
24	230	-8.03	-9.58	Lower-left
25	240	-6.25	-10.83	Lower-left
26	250	-4.28	-11.75	Lower-left

27	260	-2.17	-12.31	Lower-left
28	270	0.00	-12.50	Bottom
29	280	2.17	-12.31	Lower-right
30	290	4.28	-11.75	Lower-right
31	300	6.25	-10.83	Lower-right
32	310	8.03	-9.58	Lower-right
33	320	9.58	-8.03	Lower-right
34	330	10.83	-6.25	Lower-right
35	340	11.75	-4.28	Lower-right
36	350	12.31	-2.17	Lower-right



7. Arc Spacing

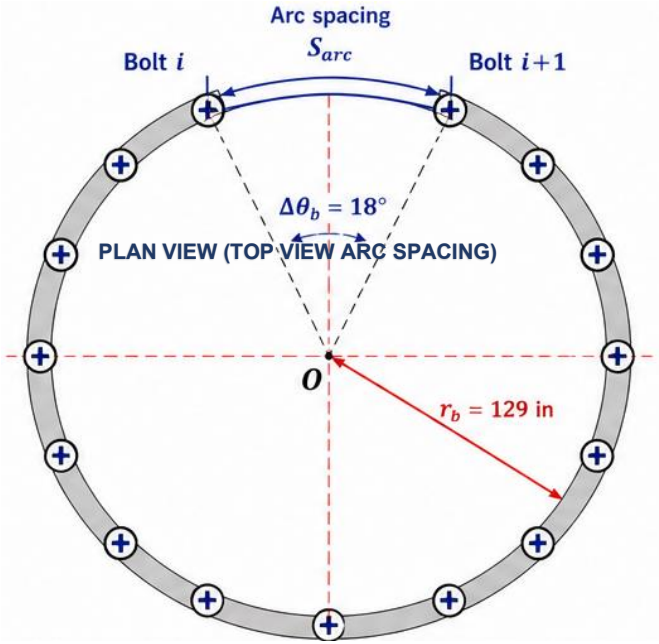
Use radius in inches:

$$r_b = 129 \text{ in}$$

$$S_{arc} = r_b \Delta\theta_b \frac{\pi}{180}$$

$$S_{arc} = 2(150)\sin 5^{\circ}$$

$$S_{arc} = 26.15 \text{ in}$$



8. Chord Spacing

$$S_{chord} = 2r_b \sin\left(\frac{\Delta\theta_b}{2}\right)$$

$$S_{chord} = 2(129)\sin(9°)$$

$$S_{chord} = 40.35 \text{ in}$$

9. Number of Bolts Per Segment

$$n_{bs} = \frac{N_b}{n_s}$$

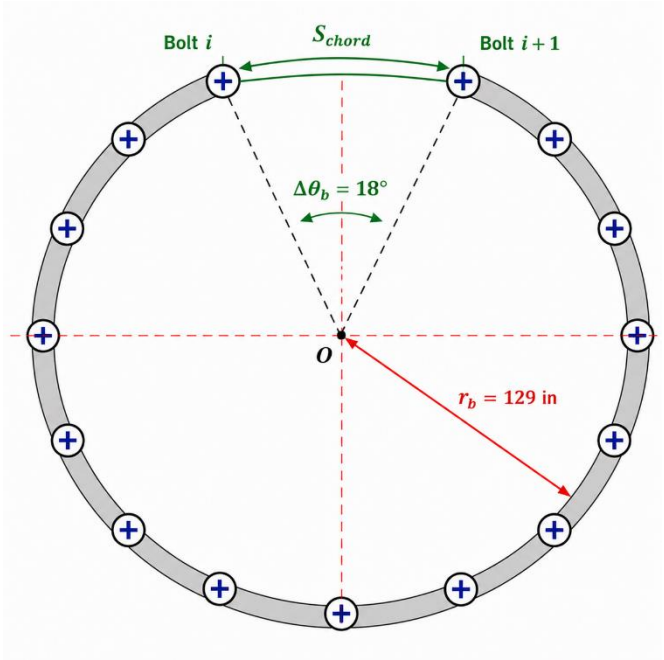
Typical choices:

- 4 segments (quadrants)
- 6 segments (60° each)
- 8 segments (45° each)

Case 1: 6 Segments

If:

- $N_b = 36$



PLAN VIEW (TOP VIEW CHORD SPACING)

- $n_s = 4$

$$\boxed{n_{bs} = 9\ bolts/segment}$$

Segment Layout

Segment	Bolts
1	1–9
2	10–18
3	19–27
4	28–36

10. Bolt Clear Edge Distance Check

If the hole is located between plate edges, the minimum clear edge distance can be written as: e=54 in

Hole diameter,

$$d_h = 1.625 \text{ in}$$

$$d_h = 1.625 \text{ in}$$

$$e_{clear} = e - \frac{d_h}{2}$$

$$e_{clear} = 54 - \frac{1.625}{2}$$

$$e_{clear} = 54 - 0.8125$$

$$e_{clear} = 53.19 \text{ in}$$

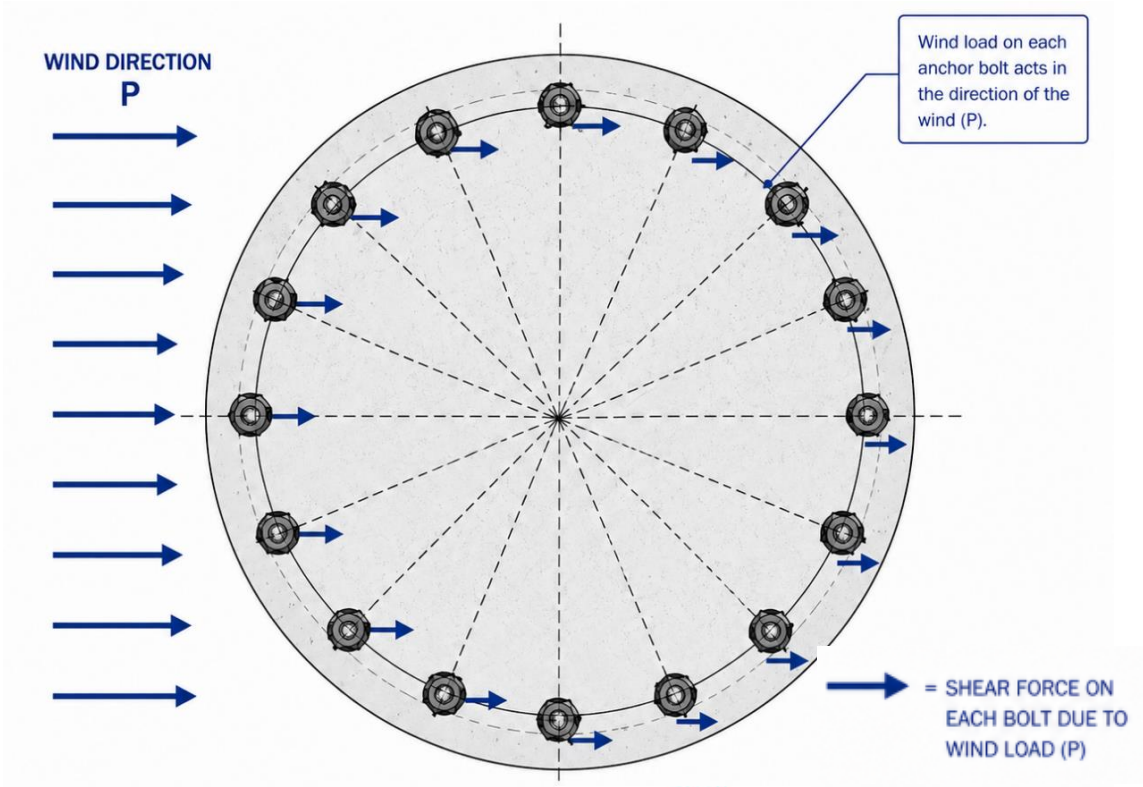
$$\boldsymbol{e_{clear} = 53.19 \text{ in}}$$

Check Against Code

From ACI / AISC / practical detailing:

$$1.5 \times d_b = 1.5 \times 1.5 = 2.25\ in$$

e_{clear}=53.19in>2.25in……✓**SAFE**



11. Shear Per Bolt

Vu=44.5 kips

$$V_{bolt} = \frac{Vu}{Nb}$$

Per bolt:

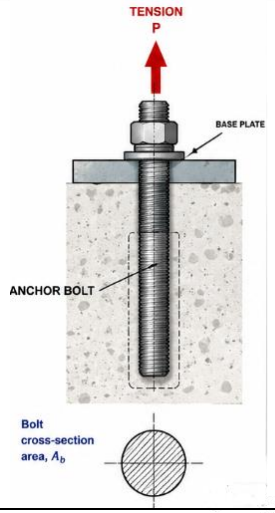
$$V_{bolt} = \frac{44.5}{36}$$

$$\boldsymbol{V_{bolt} = 1.24\ kips}$$

12. Tensile Capacity

Using:

$$\boldsymbol{\phi P_{nt} = \phi A_b F_u}$$



Given:

$\phi = 0.75$

$A_b = 1.767$

$F_u = 58$ ksi

$$\phi P_{nt} = 0.75(1.767)(58)$$

$$\boldsymbol{\phi P_{nt} = 76.87 \text{ kips}}$$

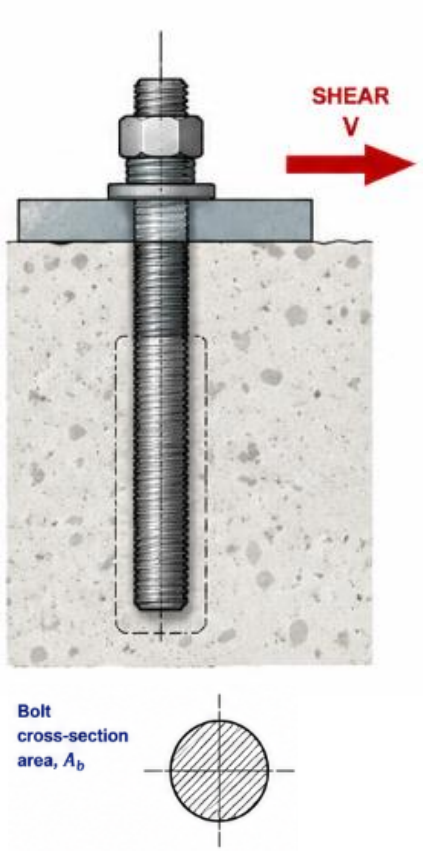
13. Shear Capacity

$$F_{nv} = 0.6F_u$$

$$F_{nv} = 0.6(58) = 34.8 \text{ ksi}$$

$$\phi V_n = 0.75(1.767)(34.8)$$

$$\boldsymbol{\phi V_n = 46.12 \text{ kips}}$$



14. Interaction Check

$$\text{Utilization} = \left(\frac{Nu}{\phi N_n}\right)^{\alpha} + \frac{V_u}{\phi V_n} \leq 1 \text{.....} \textit{ACI 318-19 Sec17.6}$$

$$\text{Utilization} = \frac{T_{bolt}}{\phi P_{nt}} + \frac{V_{bolt}}{\phi V_n} \leq 1$$

$$= \frac{18.71}{76.87} + \frac{1.24}{46.12}$$

$$= 0.243 + 0.027$$

$$= 0.270$$

$$\textit{Utilization} = 0.27 < 1.0 \Rightarrow \textit{SAFE}$$

15. Anchor Bolt Length Design:

- **Minimum embedment (code-based)**

ACI / AISC / industry:

$$h_{ef} \geq 8d_b$$

$$h_{ef} = 8 \times 1.5 = 12\ in$$

h_{ef}=25 in 25>12

16. Pull Out Test/ Concrete Breakout Capacity

$$N_b = k_c\sqrt{f_c'}\ h_{ef}^{1.5}$$

where:

- N_b = *concrete breakout strength*
- k_c = *empirical constant*
- f_c' = *concrete compressive strength*
- h_{ef} = *effective embedment depth.*
- *Embedment = approximately 9–10 ft*
- *Effective embedment = assume* $h_{ef} = 108$ *in conservative*

Based on the foundation detail drawing,

$$h_{ef} = 96\ in$$

Substituting:

$$N_b = 24\sqrt{4000}(25)^{1.5} = 24(63.25)(125)$$

$$N_b = 189736\ lb$$

$$N_b \approx 189.74\ kips$$

Using the ACI strength reduction factor:

$$\phi = 0.75$$

$$\phi N_b = 0.75(189.74)$$

$$\phi N_b \approx 142.30\ kips$$

Maximum tensile demand per anchor bolt:

$$T_{bolt} = 34.45\ kips$$

17. Concrete breakout utilization:

$$= \frac{18.71}{142.30}$$
$$= 0.131$$
$$0.131 < 1.0 \; \boldsymbol{Safe}$$

-

Parameter	Value
Arc spacing, S_{arc}	26.18 in
Chord spacing, S_{chord}	26.15 in
Bolts per segment	9
Clear edge distance	53.19 in
Shear demand per bolt	1.24 kips
Tensile capacity, ϕP_{nt}	76.87 kips
Shear capacity, ϕV_n	46.12 kips
Interaction utilization	0.27
Minimum required embedment	12 in
Provided embedment	25 in
Concrete breakout capacity, ϕN_b	142.30 kips
Concrete breakout utilization	0.131
Final Design Status	SAFE